

The entire process of coding (that the programmer prepares in Java) to the dynamic, interactive content viewed on the Web site takes place in a sequence:

1. The programmer prepares a Java application using the GWT libraries (a set of functions and routines grouped together for ease of use)
2. The GWT Java to JavaScript compiler converts the code into JavaScript and HTML files. These files usually work with all browsers, but some may require certain settings to be changed or edited.

3.2 The Code: Need to change?

The next question that arises is about the need to translate the code written in Java to be converted into JavaScript. The reason is very simple from the point of view of programming. Java offers one of the most versatile packages for the total development of applications. When we combine this power and versatility with GWT, we obtain a tool upon which the AJAX development platform thrives.

Since the GWT is compatible with all the prominent Java development tools such as Eclipse, JUnit, etc., common JavaScript errors (example: type mismatches) can be easily debugged, and therefore the result is a better product / service. Besides, Java is an object-oriented language, so it allows more work to be done with lesser documentation or detailing of steps.

3.3. Obtaining GWT For Download

The GWT SDK is a complete package for development of Web applications. It comes with a compiler, hosted Web browser, and class libraries.

To install GWT, the first and primary requirement is to have the Java SDK installed on your machine. You can get the Java SDK from <http://java.sun.com/j2se/1.4.2/download.html>. The GWT can be downloaded from code.google.com/webtoolkit/download.html.

All the GWT files are located in an extracted directory, as it does not have any installer application. The most important file for using GWT is `applicationCreator`. This file is responsible for generating batch files, support classes, and directories for the new GWT project. For a new user, Google provides some sample applications with the GWT—it is always a good idea for a newbie to familiarise oneself in uncertain waters.

3.4 GWT: Behind The Scenes

There are two modes of operation for a GWT implementation. For a user, it is interesting as well as helpful to know about these.

1. Hosted mode: here the application is run as Java bytecode within the Java Virtual Machine (JVM). The advantage of running the application in hosted mode is that since it is within the JVM, the user can utilise the debugging features of Java from within the comfort of an IDE (Integrated Development Environment).



An application in GWT

2. Web mode—the application is compiled from the Java source code to pure JavaScript and HTML with the GWT Java-to-JavaScript compiler. Therefore, the end users interact with only the Web mode version of your application.